

Lesson 14 Proof by Induction

Mathematical induction is a method for proving a statement $S(n)$ about a set of integers bounded below, $n \geq a$. There are two steps in the proof:

- **(Basis step/ Base Case)** Show that $S(a)$ is true.
- **(Inductive step)** For an arbitrary $k \geq a$, assume that $S(k)$ is true (inductive hypothesis) and show that $S(k + 1)$ is true.

With these two steps, we have shown that $S(a)$ is true, which implies $S(a + 1)$ is true, which in turn implies $S(a + 2)$ is true, etc., so $S(n)$ is true for all $n \geq a$.

Proposition 1 Prove that for every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

Proof (Basis step) The statement is true for $n = 1$:

$$1 = \frac{1(1+1)}{2}$$

(Inductive step) For an arbitrary $k \geq 1$, suppose the formula is true for $n = k$, i.e.

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}$$

Then for $n = k + 1$:

$$\begin{aligned} \underbrace{1 + 2 + \cdots + k}_{\frac{k(k+1)}{2}} + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k+1}{2}(k+2) = \frac{(k+1)((k+1)+1)}{2} \end{aligned}$$

the formula is true, and by induction, it is true for all natural numbers.

Proposition 2 For every non-negative integer n ,

$$2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$$

Proof We use induction. The statement is true for $n = 0$:

$$2^0 = 1 = 2^1 - 1 = 1$$

For an arbitrary $k \geq 0$, suppose the statement is true, then the formula is true for $n = k + 1$:

$$\begin{aligned} \underbrace{2^0 + 2^1 + 2^2 + \cdots + 2^k}_{2^{k+1} - 1} + 2^{k+1} &= (2^{k+1} - 1) + 2^{k+1} \\ &= 2(2^{k+1}) - 1 \\ &= 2^{k+2} - 1 \\ &= 2^{(k+1)+1} - 1 \end{aligned}$$

Proposition 3 For every natural number n ,

$$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1$$

Proof The statement is true for $n = 1$:

$$1 \cdot 1! = (1+1)! - 1 = 1$$

For an arbitrary $k \geq 1$, suppose the statement is true:

$$1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! = (k+1)! - 1$$

We will show that the statement is true for $n = k + 1$:

$$\begin{aligned} & 1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! + (k+1) \cdot (k+1)! \\ &= (k+1)! - 1 + (k+1) \cdot (k+1)! \\ &= (k+1)! + (k+1) \cdot (k+1)! - 1 \\ &= (k+1)! (1 + (k+1)) - 1 \\ &= (k+1)! (k+2) - 1 \\ &= (k+2)! - 1 \end{aligned}$$

Proposition 4 For all natural numbers n , $n^3 - n$ is divisible by 3.

Proof The statement is true for $n = 1$:

$$1^3 - 1 = 0 = 3 \cdot 0$$

For an arbitrary $k \geq 1$, if $k^3 - k = 3m$ for some $m \in \mathbb{Z}$, then for $n = k + 1$,

$$\begin{aligned} (k+1)^3 - (k+1) &= k^3 + 3k^2 + 3k + 1 - k - 1 \\ &= (k^3 - k) + 3(k^2 + k) \\ &= 3m + 3(k^2 + k) \\ &= 3(m + k^2 + k) \end{aligned}$$

Since $m + k^2 + k$ is an integer, the statement is true for $n = k + 1$.

Proposition 5 Consider the recursive sequence a_n :

$$a_1 = 3, \quad a_n = 2a_{n-1} - n + 2 \text{ for } n \geq 2$$

Then $a_n = 2^n + n$ for all $n \in \mathbb{N}$.

Proof The statement is true for the base case $n = 1$,

$$a_1 = 3 = 2^1 + 1$$

Let $k \geq 1$ be a natural number. Suppose that $a_k = 2^k + k$, then

$$a_{k+1} = 2a_k - (k+1) + 2 = 2(2^k + k) - k + 1 = 2^{k+1} + (k+1)$$

Proposition 6 Let $n \in \mathbb{N}$. A $2^n \times 2^n$ chess board with one square removed can be tiled by L-triominoes.

Proof The proposition is true for $n = 1$, i.e. a 2×2 board:



If $n = k + 1$, the $2^{k+1} \times 2^{k+1} = 2(2^k) \times 2(2^k)$ board with one square removed may be divided into 4 quarters. We can put a L-triomino in the center such that each quarter is a $2^k \times 2^k$ board with one square removed, as shown. Then by the induction hypothesis, each $2^k \times 2^k$ board can be tiled by L-triominoes.

So, the entire $2^{k+1} \times 2^{k+1}$ board with one square removed can be tiled by L-triominoes.

